Deep Learning

Foundations: Neural Networks to Representation Learning

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Outline

1. Deep Learning Foundations
2. Convolutional Neural Networks
3. Sequence Models & Transformers
4. Unsupervised & Representation Learning





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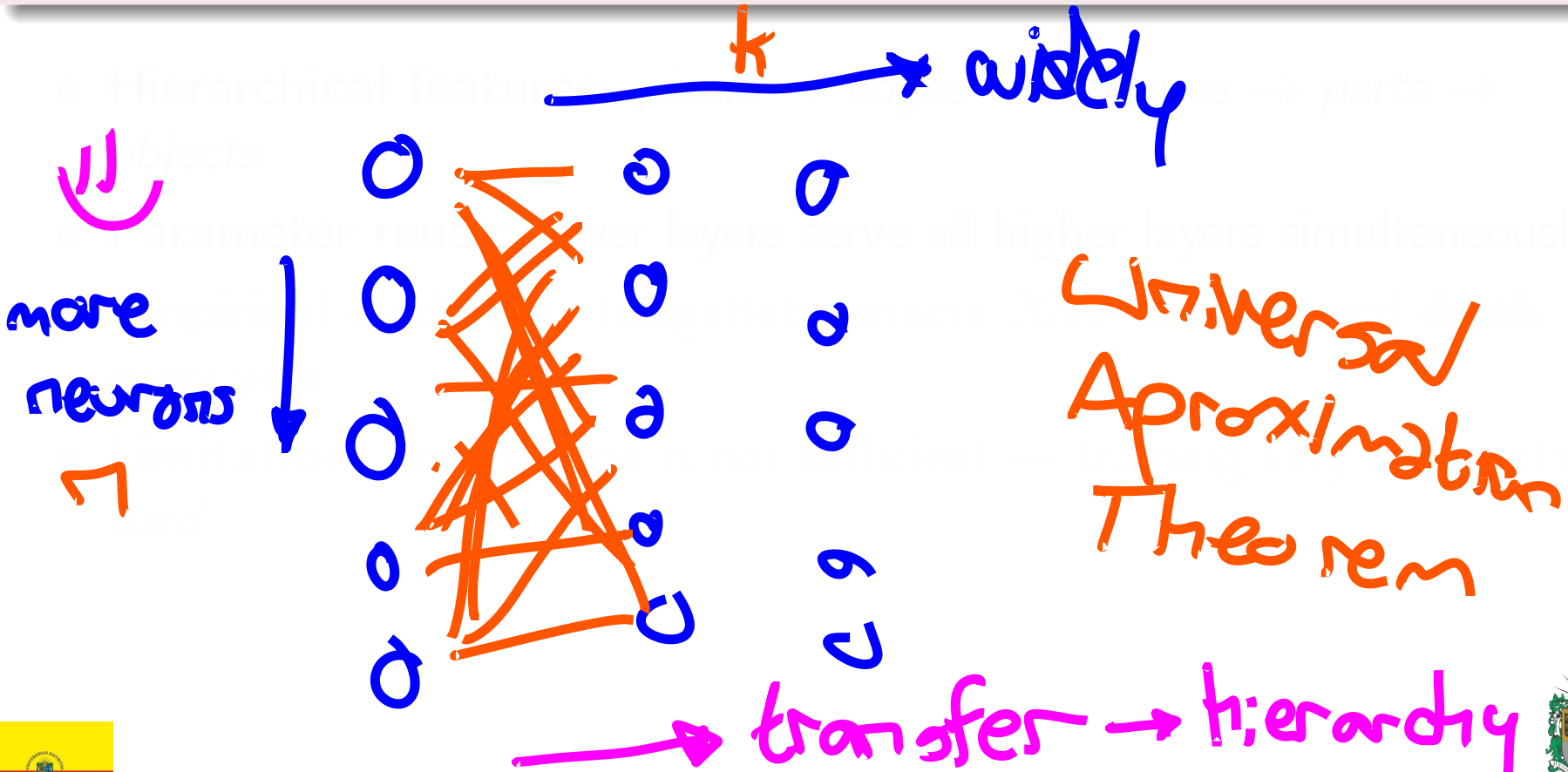




Why Depth?

Exponential Representational Efficiency

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- **Parameter reuse**: lower layers serve all higher layers simultaneously
- **Empirical evidence**: ImageNet winners *2012–2016* gained depth every year
- **Limitation**: depth alone is not sufficient; training very deep nets is *hard*





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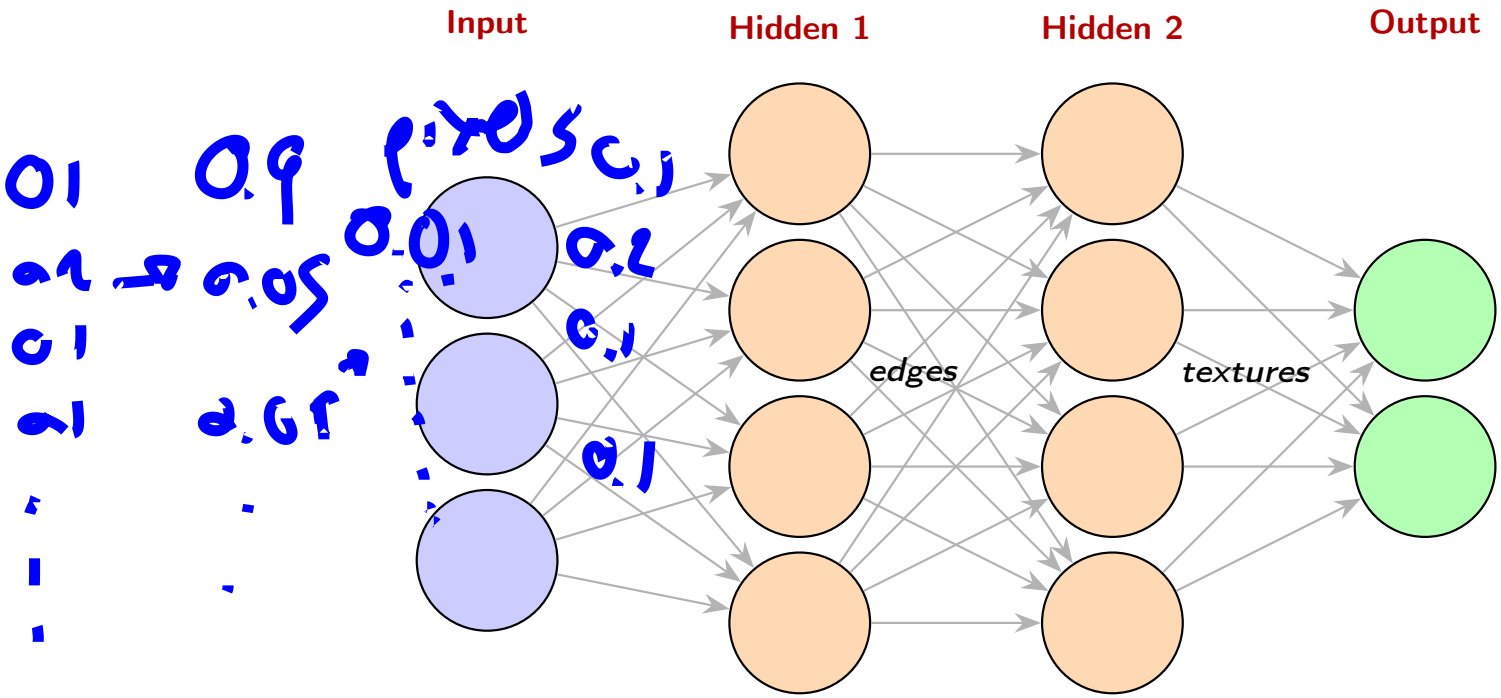
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Neural Network Depth: Visual Intuition

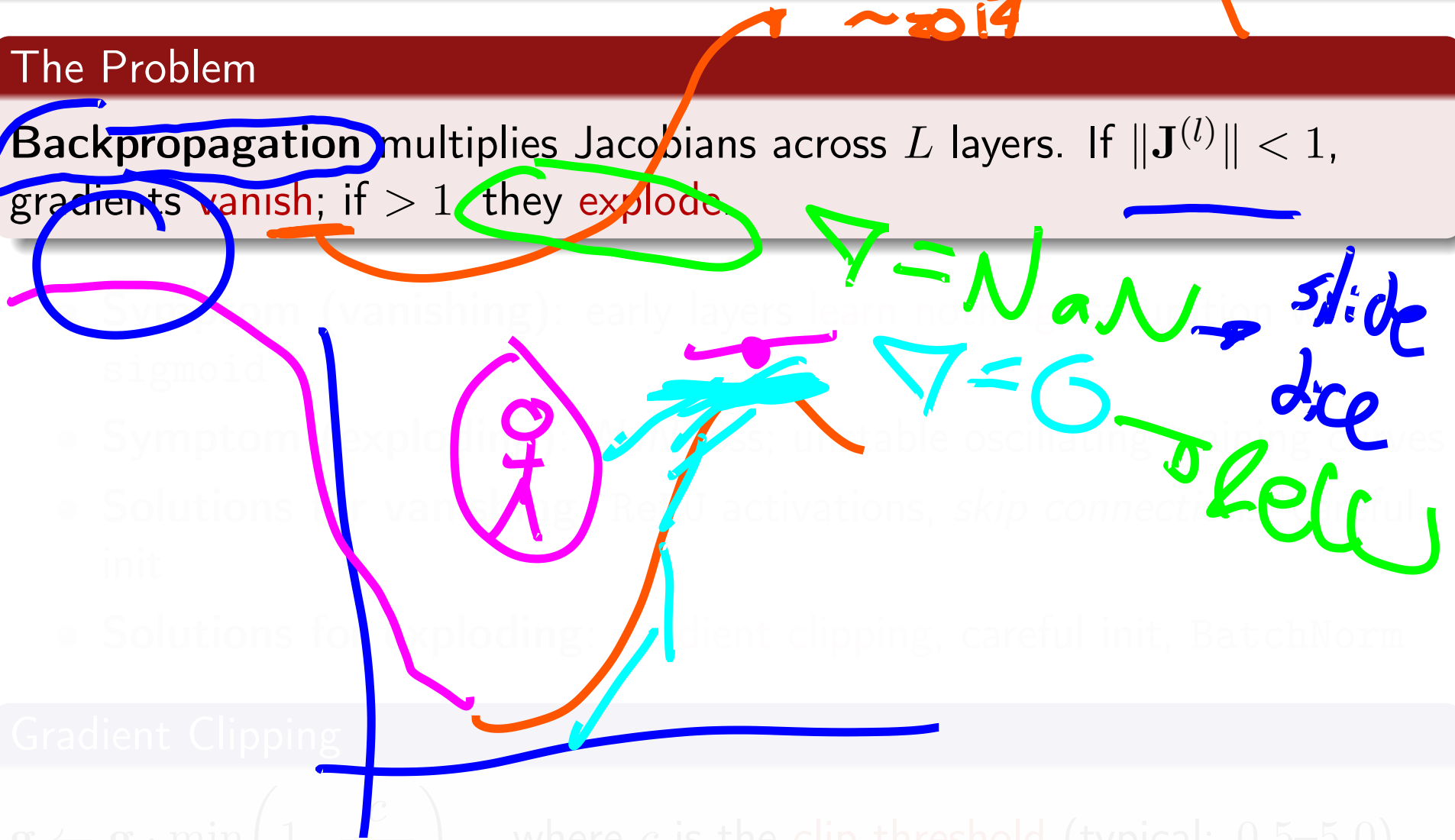


Each layer learns more abstract representations.

Vanishing and Exploding Gradients

The Problem

Backpropagation multiplies Jacobians across L layers. If $\|\mathbf{J}^{(l)}\| < 1$, gradients vanish; if > 1, they explode.



Gradient Clipping

$$\mathbf{g} \leftarrow \mathbf{g} \cdot \min\left(1, \frac{c}{\|\mathbf{g}\|}\right)$$ where c is the clip threshold (typical: 0.5–5.0)

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- **Symptom (exploding)**: *NaN* loss; unstable oscillating training curves
- Solutions for vanishing: `ReLU` activations, *skip connections*, careful init
- Solutions for exploding: gradient clipping, careful init, `BatchNorm`

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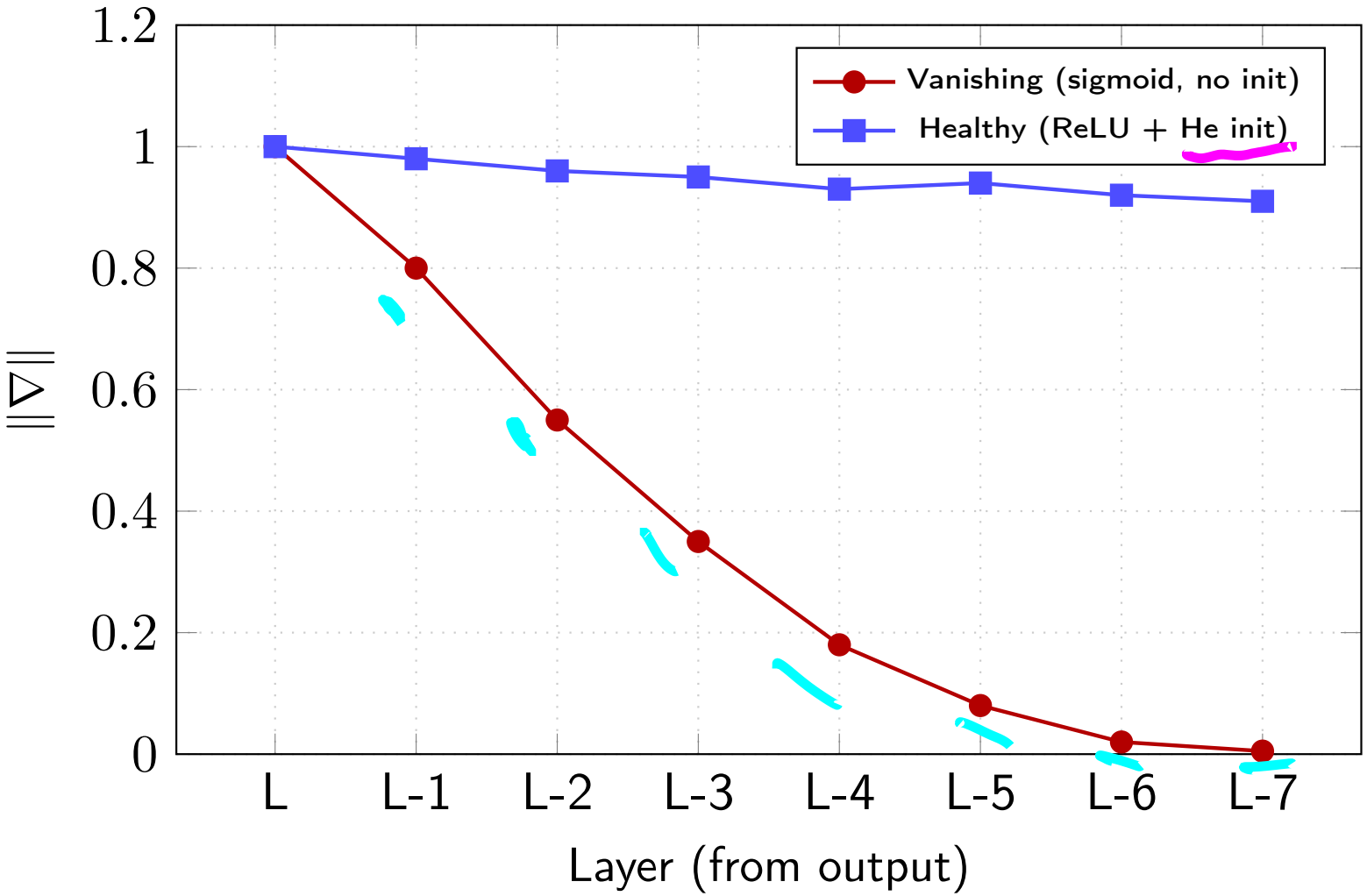
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Gradient Magnitude Across Layers



Vanishing: gradients collapse to zero far from the output.

Weight Initialization

`Xavier / Glorot`

$$W \sim \mathcal{N}\left(0,\ \frac{2}{n_{\text{in}} + n_{\text{out}}}\right)$$

- Designed for `tanh` / `sigmoid`
- Keeps variance stable through layers

`He / Kaiming`

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- Accounts for *half the units* being zero
- Default for modern networks

Zero Init Kills Learning

If all weights are *zero*, all neurons compute identical gradients — *symmetry* never breaks. Always break symmetry with *random initialisation*.





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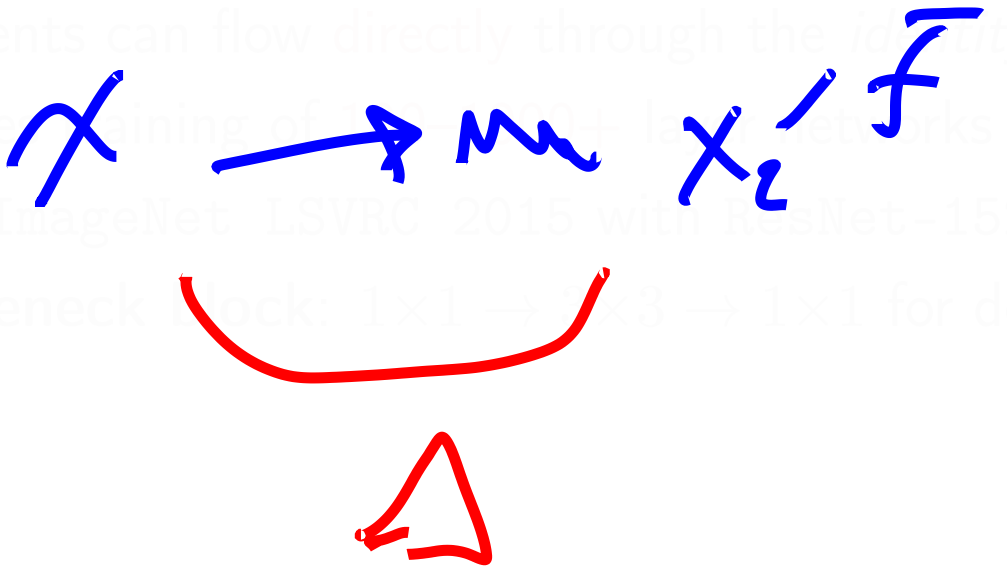
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Skip Connection (He *et al.*, 2016)

Instead of learning $\mathcal{H}(\mathbf{x})$, learn the residual:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \mathbf{W}) + \mathbf{x}$$

where $\mathcal{F}$ is a stack of conv + BN + ReLU layers.



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where $\mathcal{F}$ is a stack of conv + BN + ReLU layers.

- Gradients can flow directly through the *identity shortcut*
- Enables training of 100–1000+ layer networks
- Won ImageNet LSVRC 2015 with ResNet-152 (3.57% top-5 error)
- **Bottleneck block**: $1\times1 \rightarrow 3\times3 \rightarrow 1\times1$ for deeper/efficient variants

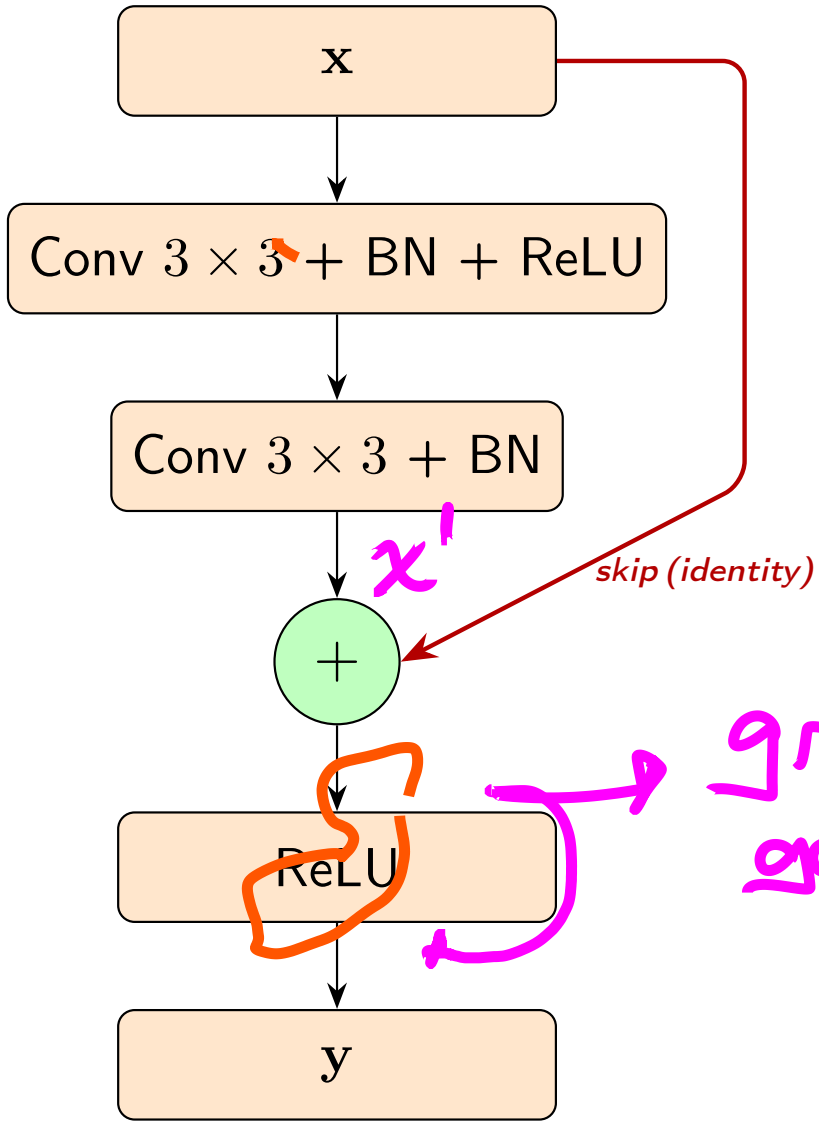
great contribution





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ResNet Skip Connection: Block Diagram



gradient on good conditions

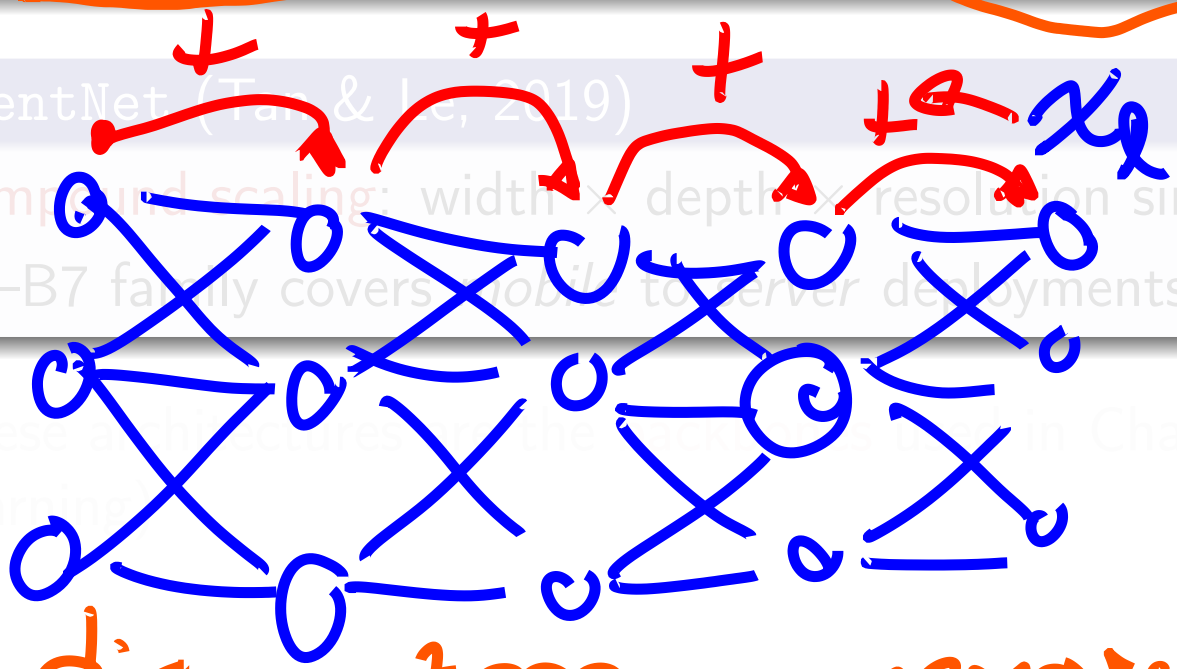
Other Deep Architectures

DenseNet (Huang *et al.*, 2017)

Each layer receives feature maps from all preceding layers:

$$\mathbf{x}_l = \mathcal{H}_l([\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{l-1}])$$

- Maximum gradient flow; strong feature reuse
- Fewer parameters than ResNet at similar accuracy



disadvantage → memory

EfficientNet (Tan & Le, 2019)

- Compound scaling: width × depth × resolution simultaneously
- B0–B7 family covers *mobile* to *server* deployments
- These architectures won the ILSVRC Image Classification Challenge (Transfer Learning)

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Normalization Variants

Batch Normalization (BN)

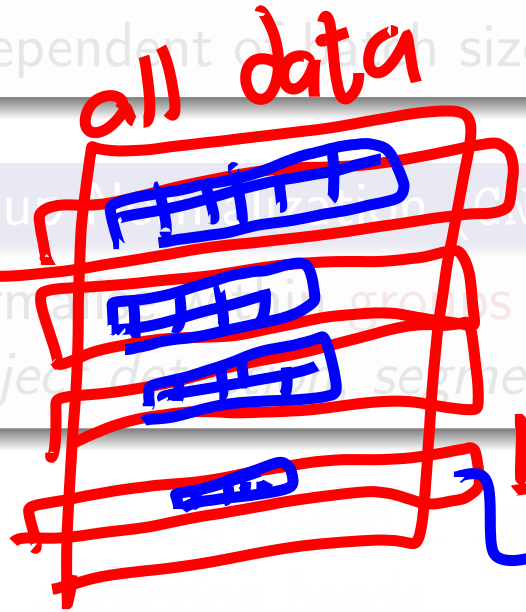
Normalise over the batch dimension ideal for CNNs with large batches

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Normalise over the feature dimension; ideal for Transformers and RNNs; independent of batch size

Group Normalization (GN)

Normalise over groups of channels; better than BN for small batches (*object detection, segmentation*)



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time series

BERT ↔ GPT

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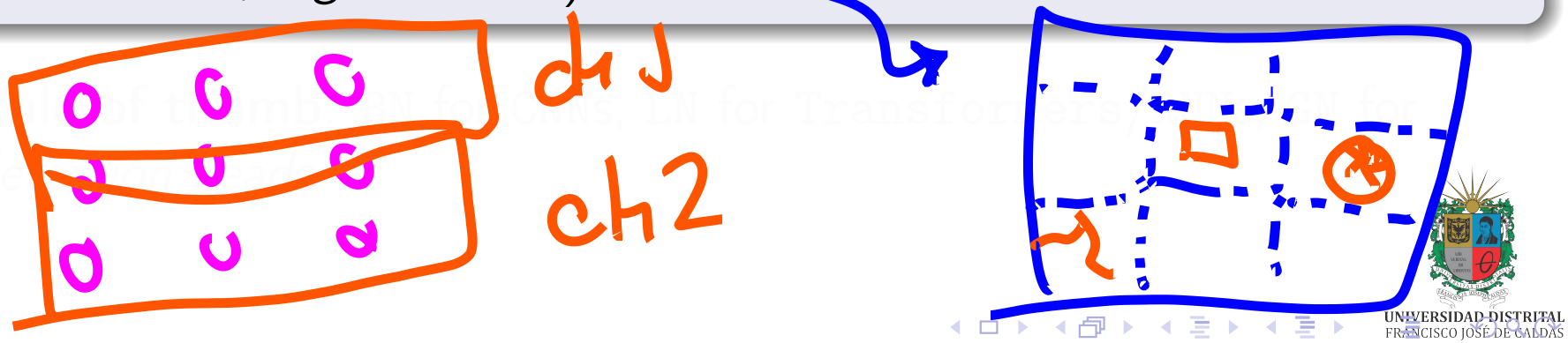
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Practical Example: Training Deep vs Shallow Networks

- **Task**: CIFAR-10 image classification
- **Model A**: plain 20-layer CNN (no skip connections)
- **Model B**: ResNet-20 (same depth, with skip connections)
- **Observation**: Model A shows higher training error than a shallower network (degradation problem); ResNet-20 avoids this
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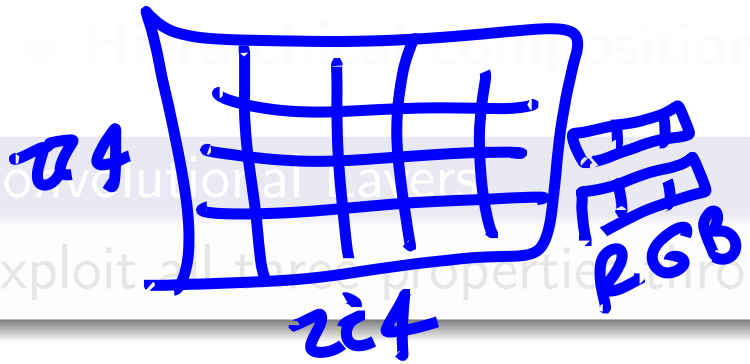


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Motivation: Images as Data

The Problem with Plain MLPs

A $224 \times 224 \times 3$ image has $150,528$ input dimensions. A single *fully-connected* layer with $1,000$ neurons requires $150M$ parameters.





Convolutional Layers

Exploit all these properties through weight sharing and local connectivity.

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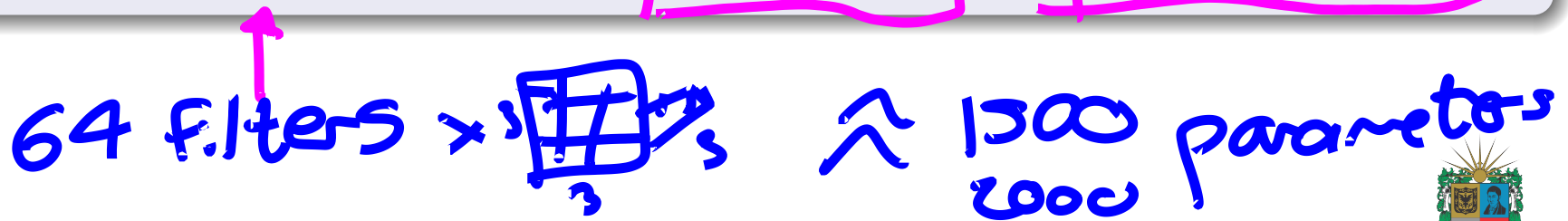
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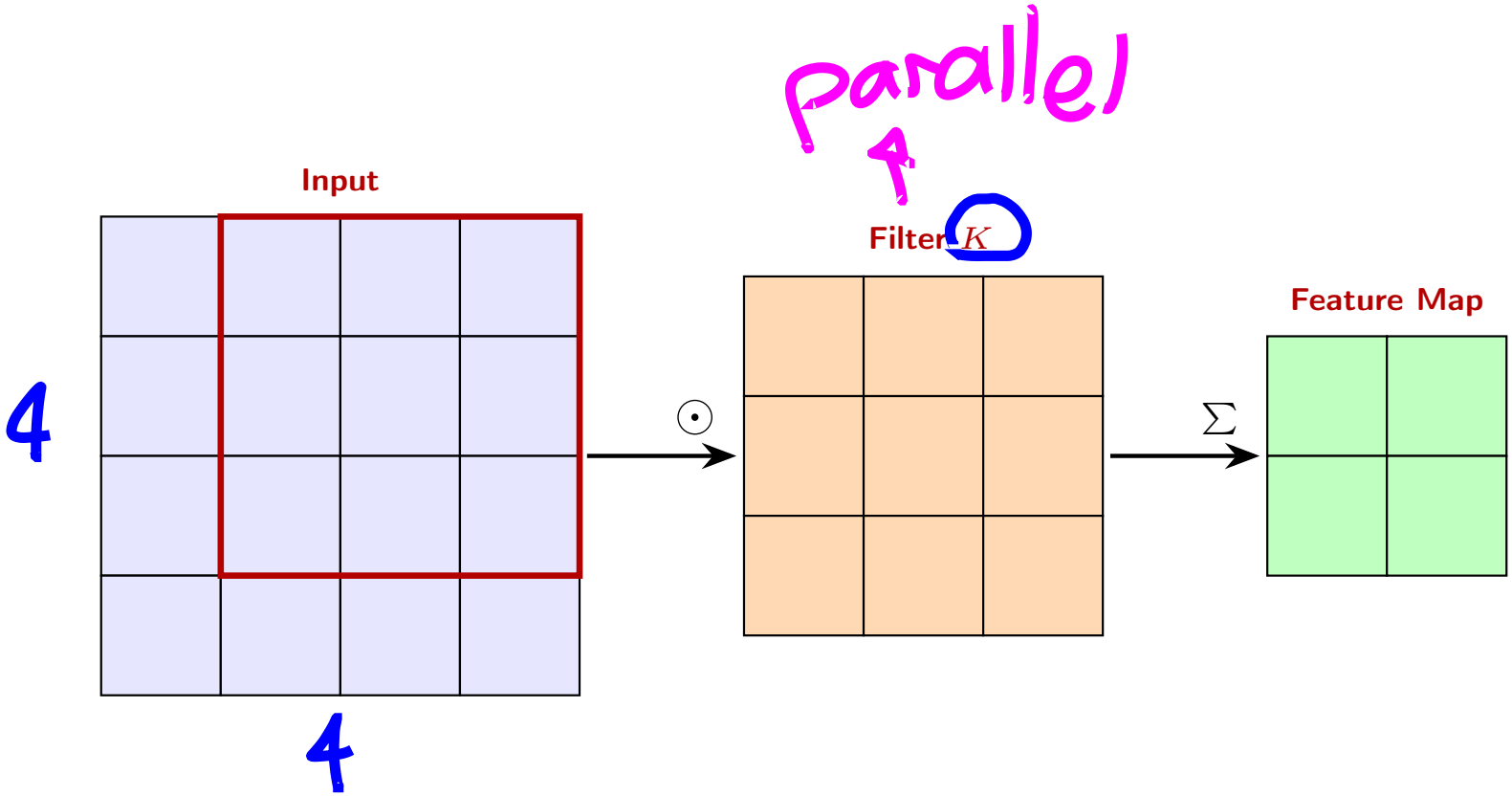
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Convolution Operation: Visual Diagram



Sliding the filter over the input with *stride* S produces the *feature map*.



The Convolutional Layer

Convolution Equation

For input $\mathbf{I}$, kernel $\mathbf{K}$, padding P, stride S:

$$\text{out}(i,j) = \sum_{a=0}^{F-1} \sum_{b=0}^{F-1} \mathbf{K}(a,b) \cdot \mathbf{I}(i \cdot S + a, j \cdot S + b)$$

The result $\mathbf{K}(a,b)$ weights the input patch element-wise.

feature map

$$\frac{W - F + 2P}{S} + 1$$

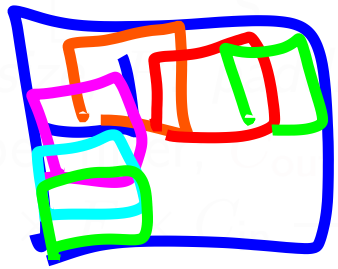


image 28×28

F 3×3

P 1

S 1 → S=2

28×28 → 14×14

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- Output (feature map) size: $\left\lfloor \frac{W - F + 2P}{S} \right\rfloor + 1$
- W: input width, F: *filter size*, P: *padding*, S: *stride*
- **Depth**: one feature map per filter, C_{out} filters in total
- **Parameters per filter**: $F \times F \times C_{\text{in}}$ — independent of W

Key Insight: *Weight Sharing*

The same filter detects the same pattern everywhere in the image.
Parameters count drops from 150M to thousands.

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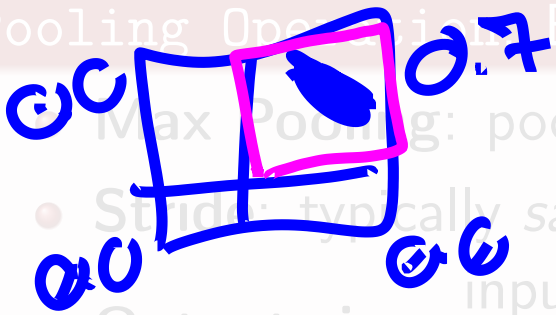
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Pooling Layers

Max Pooling

- Takes maximum in 2×2 window
- Preserves strongest activation
- Provides translation invariance
- Standard in classification nets



Average Pooling

- Takes mean in window
- Smoother; used in deeper nets
- **Global Average Pooling (GAP)**: collapse entire feature map to one value per channel
- Replaces FC layers in modern nets



Pooling Operation Example

- Max Pooling: $\text{pool}(a, b, c, d) = \max(a, b, c, d)$ on 2×2 window
- Stride: typically *same as pool size* to avoid overlap
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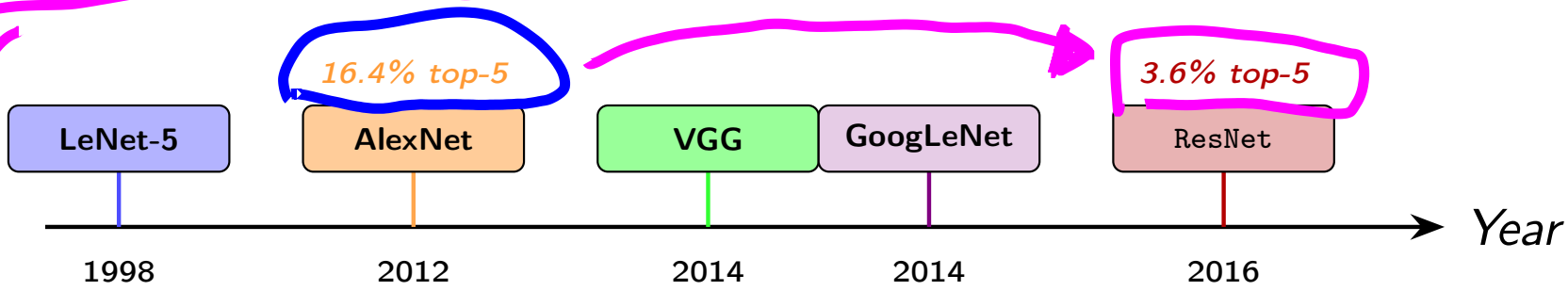
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Classic CNN Architectures

- LeNet-5 (*LeCun*, 1998): first practical CNN for digit recognition
- AlexNet (*Krizhevsky*, 2012): ImageNet breakthrough; ReLU, Dropout, *GPU training*
- VGG-16/19 (*Simonyan*, 2014): stacked 3×3 *convolutions*; very deep and clean design
- GoogLeNet/Inception (*Szegedy*, 2014): Inception modules with 1×1 convolutions
- ResNet (*He*, 2016): skip connections; 152 layers; still a strong baseline



Each milestone brought lower error and deeper networks

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Why It Matters

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- CutMix: pastes random patch from one image into another
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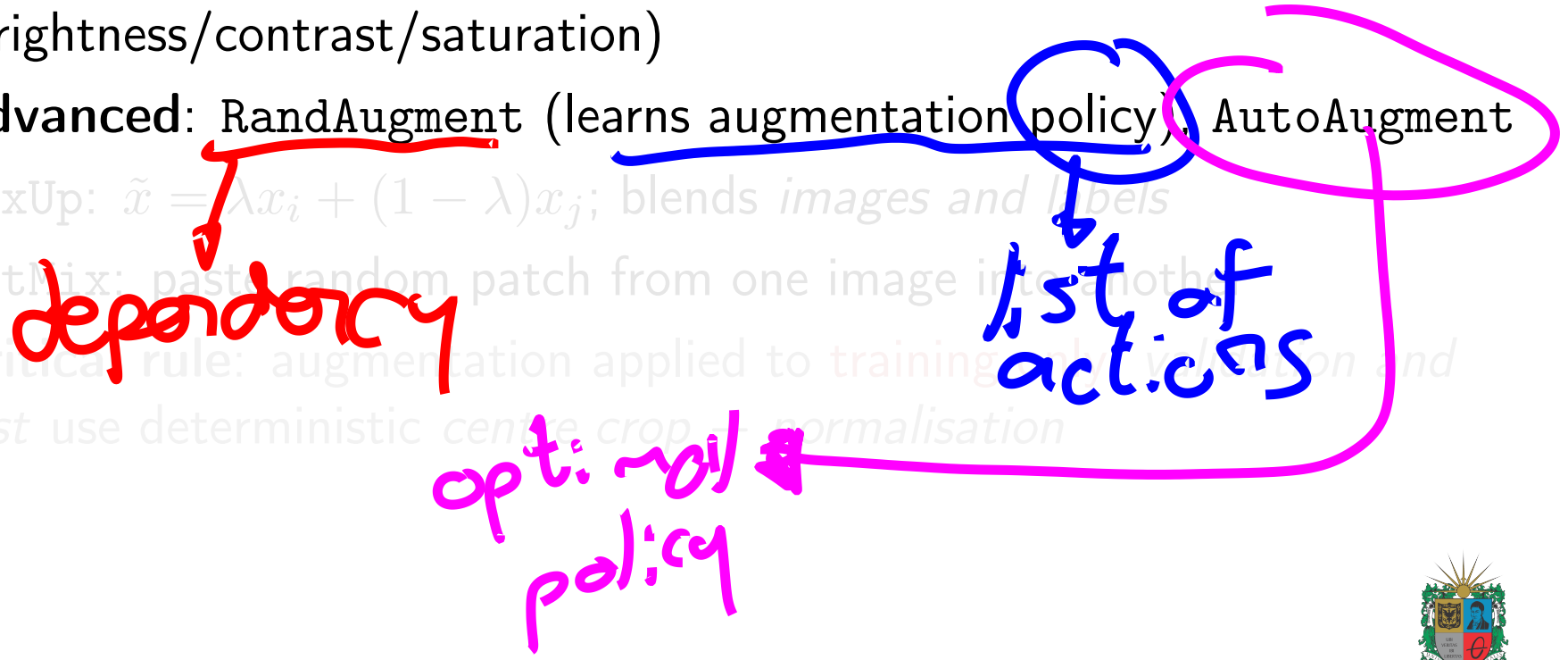


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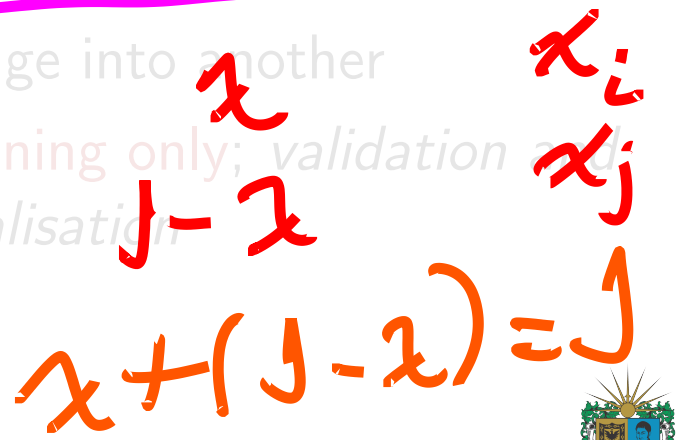
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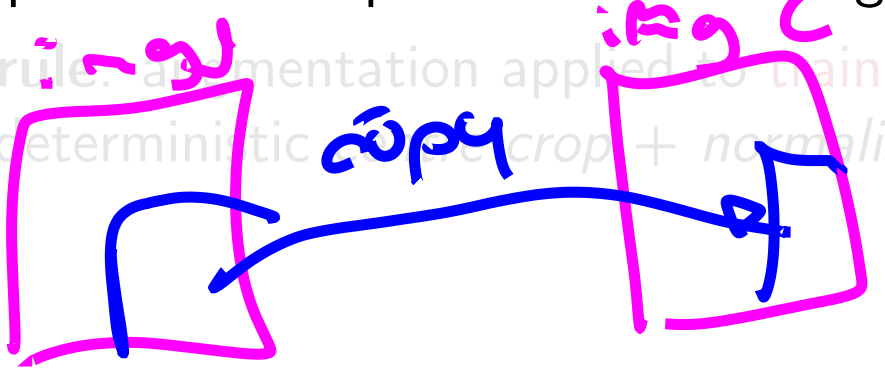


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Receptive Field and Depth

Receptive Field

The region of the input that influences a given output neuron. **Deep networks** accumulate large receptive fields without large kernels.

- Two stacked 3×3 *conv* layers have the same receptive field as one 5×5 but with fewer parameters and two nonlinearities.
- VGG insight: stack small kernels instead of using large ones.
- Dilated (*atrous*) convolutions: expand receptive field without stride.
- Depthwise separable convolutions: MobileNet family; reduce parameters $8\times$.





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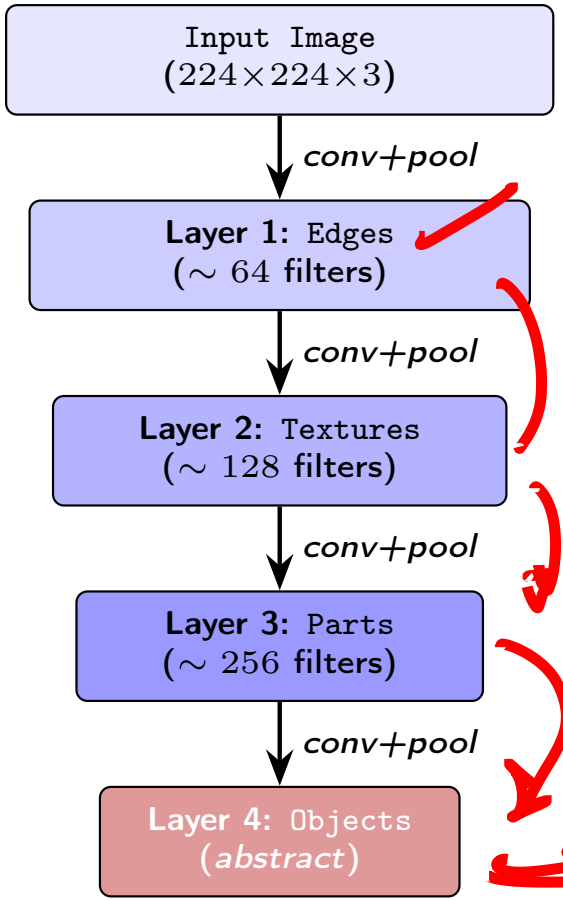
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CNN *Feature Hierarchy*: Visual Progression



Increasing abstraction, decreasing spatial size, increasing semantic meaning

Each CNN layer learns hierarchical features building towards classification

Practical Example: Training Deep vs Shallow Networks (CNN Edition)

- **Dataset**: `CIFAR-10` ($60,000$ images, 32×32, 10 classes)
- **Baseline**: simple *3-layer* `CNN` (no tricks)
- **Experiment A**: add `BatchNorm` and `Dropout`
- **Experiment B**: replace plain `CNN` with `ResNet-20`
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Outline

1 Deep Learning Foundations

2 Convolutional Neural Networks

3 Sequence Models & Transformers

4 Unsupervised & Representation Learning





Why Sequential Models?

The Temporal Dimension

Many *real-world* data sources are ordered sequences where context matters.

temporal

CNN → spatial

Solution Landscape

RNN/LSTM/GRU → Attention → Transformer → *Foundation Models* (BERT, GPT)



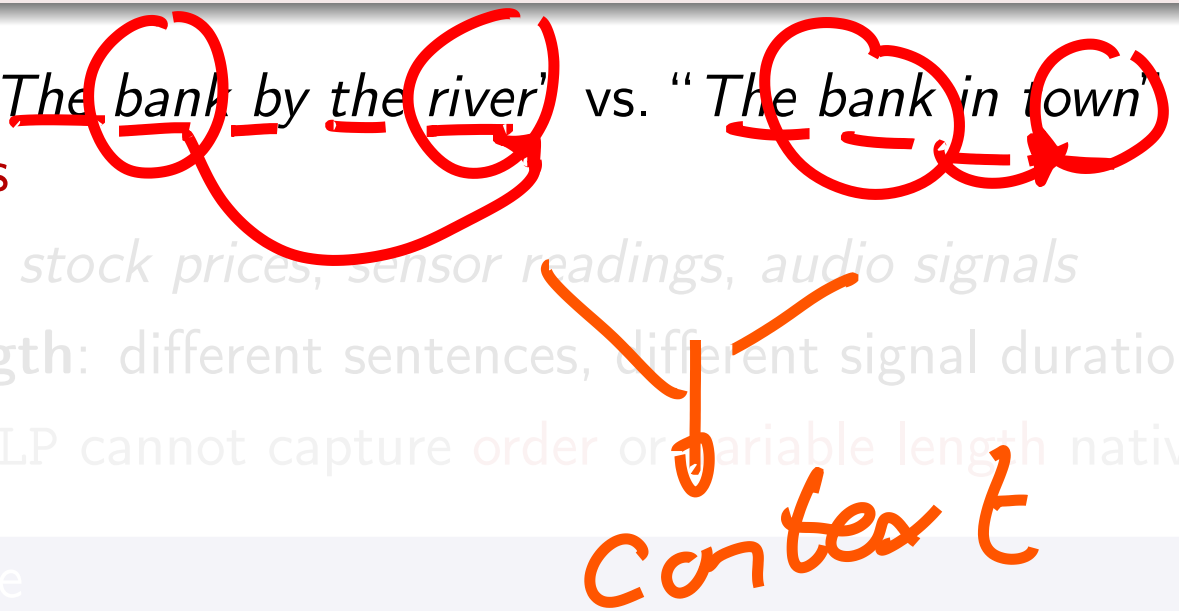


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- **Language**: "*The bank by the river*" vs. "*The bank in town*" — order disambiguates
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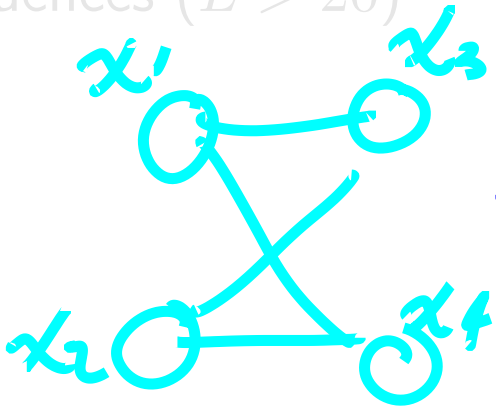


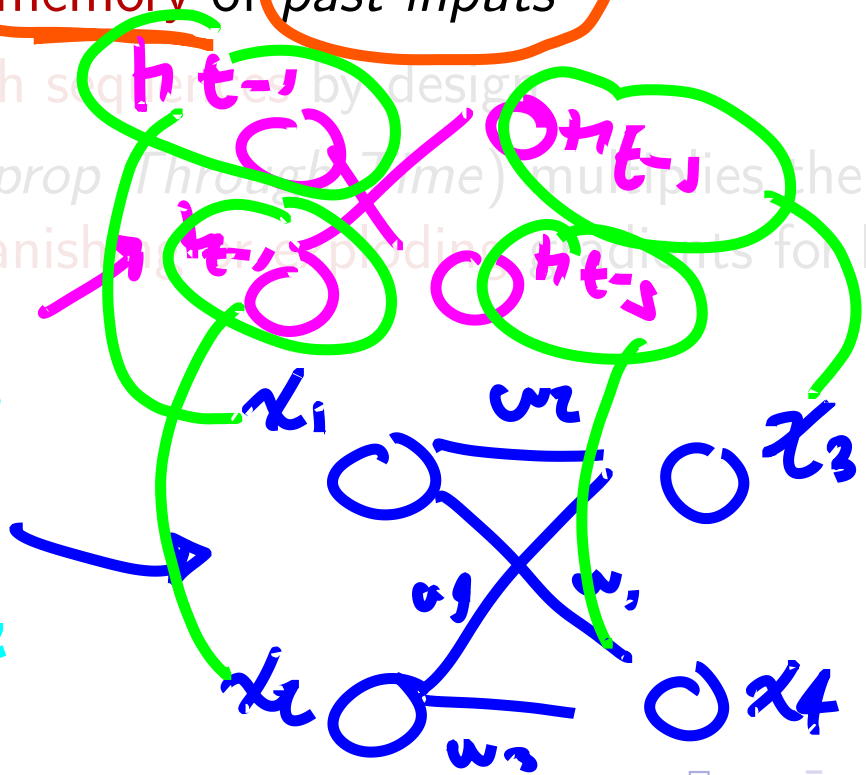
Recurrent Neural Networks (RNN)

Recurrent Update

$$\mathbf{h}_t = \tanh(\mathbf{W}_{xh}\mathbf{x}_t + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h)$$

- Weight sharing across time: same $\mathbf{W}_{xh}$, $\mathbf{W}_{hh}$ for *every step*
- Hidden state $\mathbf{h}_t$ is the memory of *past inputs*
- Supports variable-length sequences by design
- Problem: BPTT (*Backprop Through Time*) multiplies the same Jacobian L times $\rightarrow$ vanishing/exploding gradients for long sequences ($L > 20$)





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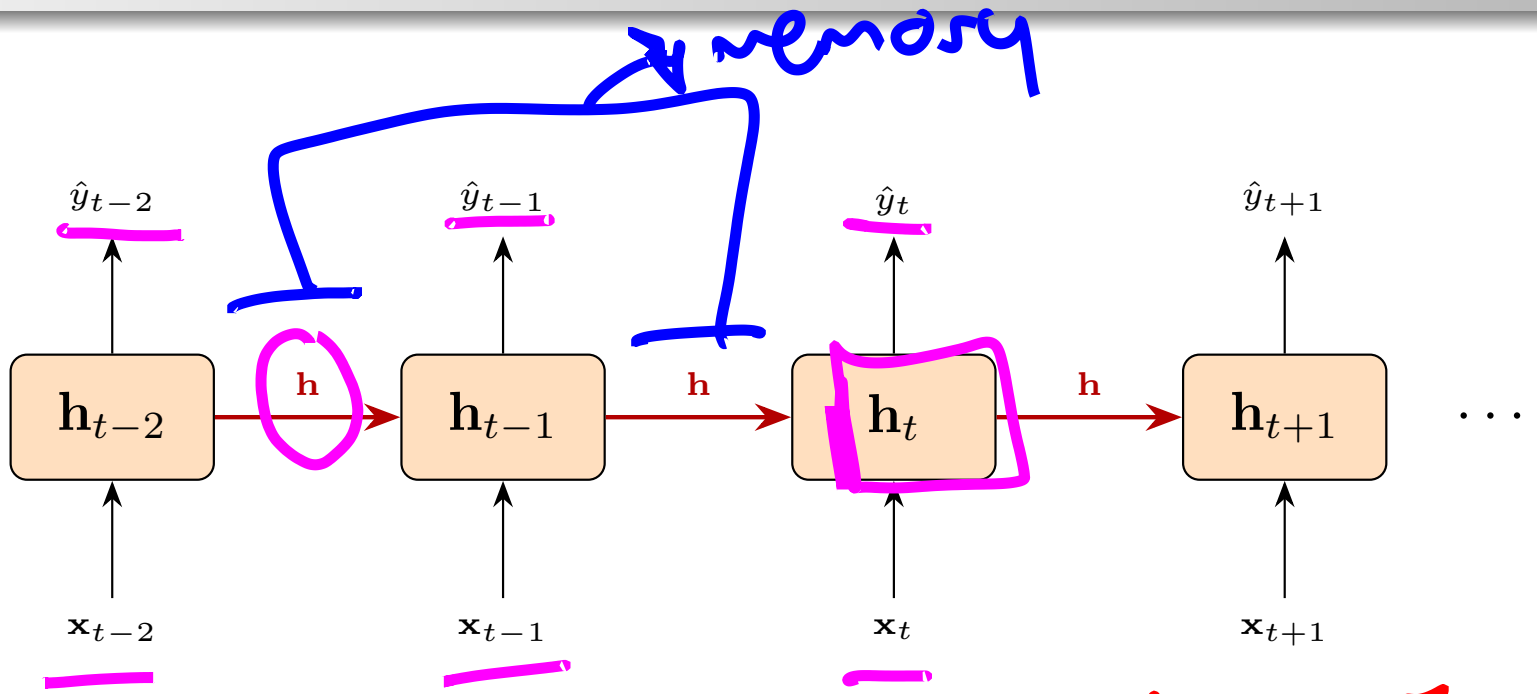
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RNN Unrolled: Information Flow



initial → no learning

The *Backpropagation Through Time* (BPTT) Problem

As gradients flow backward through L time steps, they are multiplied by the recurrent Jacobian L times. This causes exponential scaling.

Tanh → $|J| < 1$





Long Short-Term Memory (*LSTM*)

Gated Cell (Hochreiter & Schmidhuber, 1997)

- **Forget gate**: $\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f)$
- **Input gate**: $\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i)$
- **Cell state**: $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$

- Cell state $\mathbf{c}_t$: *long-term memory*; gradient highway
- Hidden state $\mathbf{h}_t$: *short-term*; passed to the next step and the output
- **Forget gate**: what to erase from memory
- GRU (Cho *et al.*, 2014): simplified LSTM with *two gates* — similar performance, fewer parameters

Core idea

LSTM is a controlled memory cell: it learns when to *remember*, when to *forget*, and when to *expose state*.

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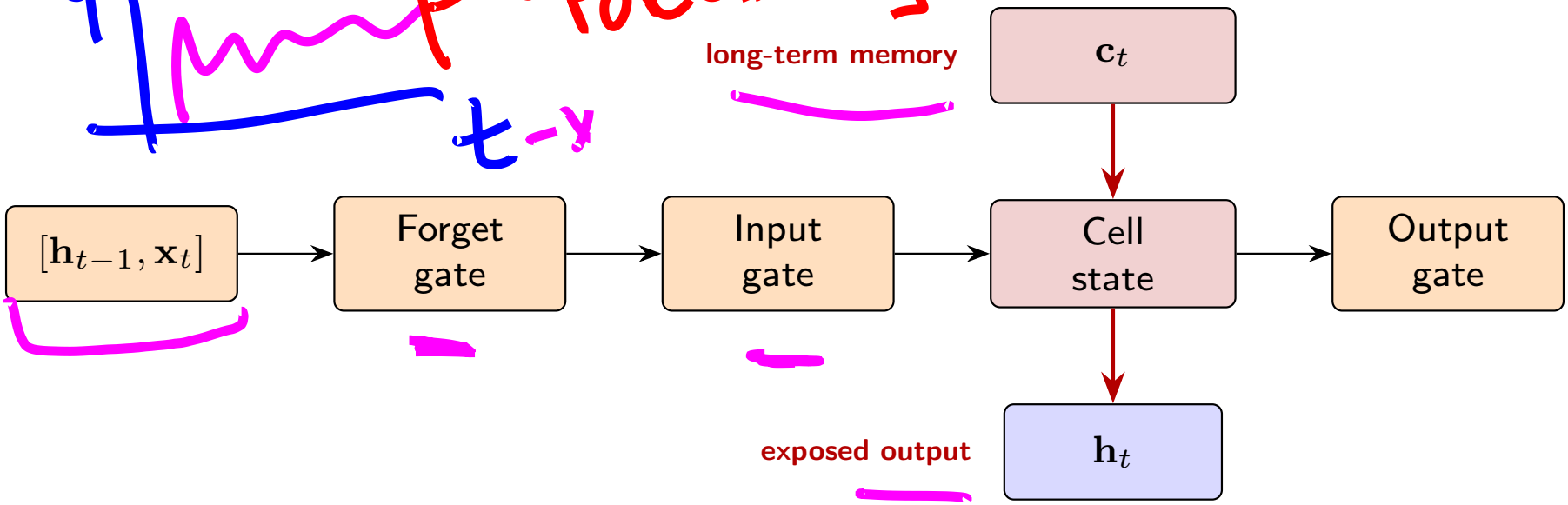
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LSTM Cell: Gate Diagram



Intuition

The *forget gate* erases outdated information, the *input gate* writes new information, the *cell state* acts as a stable memory highway, and the *output gate* exposes selected information to the next step.

LSTM vs GRU: Trade-offs

LSTM (3 gates)

- More expressive: separate *cell state* and *hidden state*
- Slower and *more parameters* ($\sim 4\times$ RNN)
- **Better** for very long dependencies ($100+$ steps)
- Standard choice for *sequence-to-sequence* and *language modelling*

GRU (2 gates)

- Simpler: combines cell and hidden states into single state
- Faster and *fewer parameters* ($\sim 3\times$ RNN)
- Similar *performance* in practice; often preferred for *efficiency*
- Easier to understand and debug than LSTM





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Practical Choice: *When to Use What*

Decision Framework

- **Short sequences** (< 20 steps), *efficiency matters*: `GRU` or even simple `RNN`
- **Long sequences** (> 50 steps), *capture complex patterns*: `LSTM`
- **Language tasks** (translation, QA, generation): `LSTM` or `Transformer`
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Modern Trend

`Transformers` have largely replaced `LSTM`/`GRU` for NLP tasks due to parallelism, but `LSTMs` remain popular for *time series* and resource-constrained settings.





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Attention Mechanism

Attention (Bahdanau *et al.*, 2015)

Let the **decoder** query all *encoder hidden states* $\mathbf{h}_j$ simultaneously:

$$e_{ij} = \text{score}(\mathbf{s}_i, \mathbf{h}_j), \quad \alpha_{ij} = \frac{\exp(e_{ij})}{\sum_k \exp(e_{ik})}, \quad \mathbf{c}_i = \sum_j \alpha_{ij} \mathbf{h}_j$$

- α_{ij}: soft alignment weight — how much position i *attends to* j
- Abolishes the information bottleneck of the fixed-size context vector
- Made *neural machine translation* practical ($\gg 10$ words)
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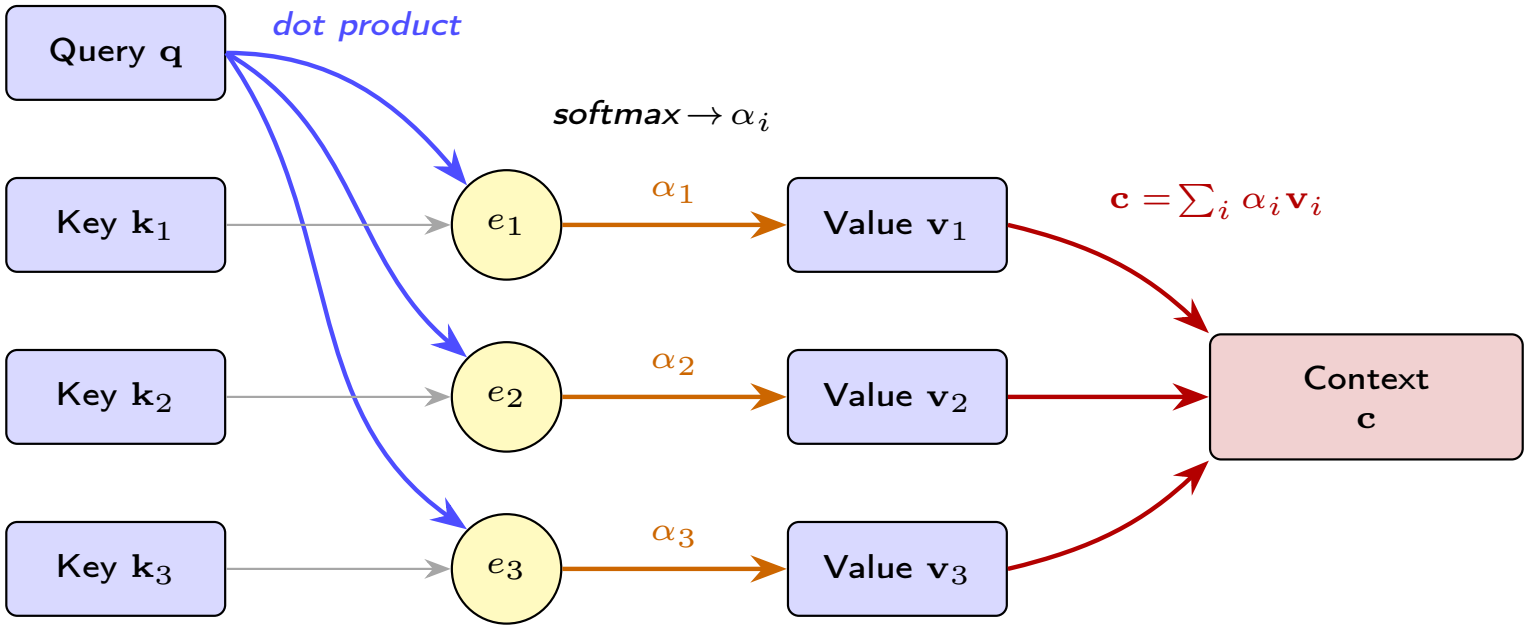
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Attention Mechanism: Vector View



The query determines which values contribute most to the context vector.

The Transformer

`Transformer` (Vaswani *et al.*, 2017) — "Attention Is All You Need"

No recurrence; entirely based on *self-attention* and *feed-forward* layers.

Scaled Dot-Product Self-Attention

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}$$

- $\mathbf{Q} = \mathbf{X}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{X}\mathbf{W}_K$, $\mathbf{V} = \mathbf{X}\mathbf{W}_V$
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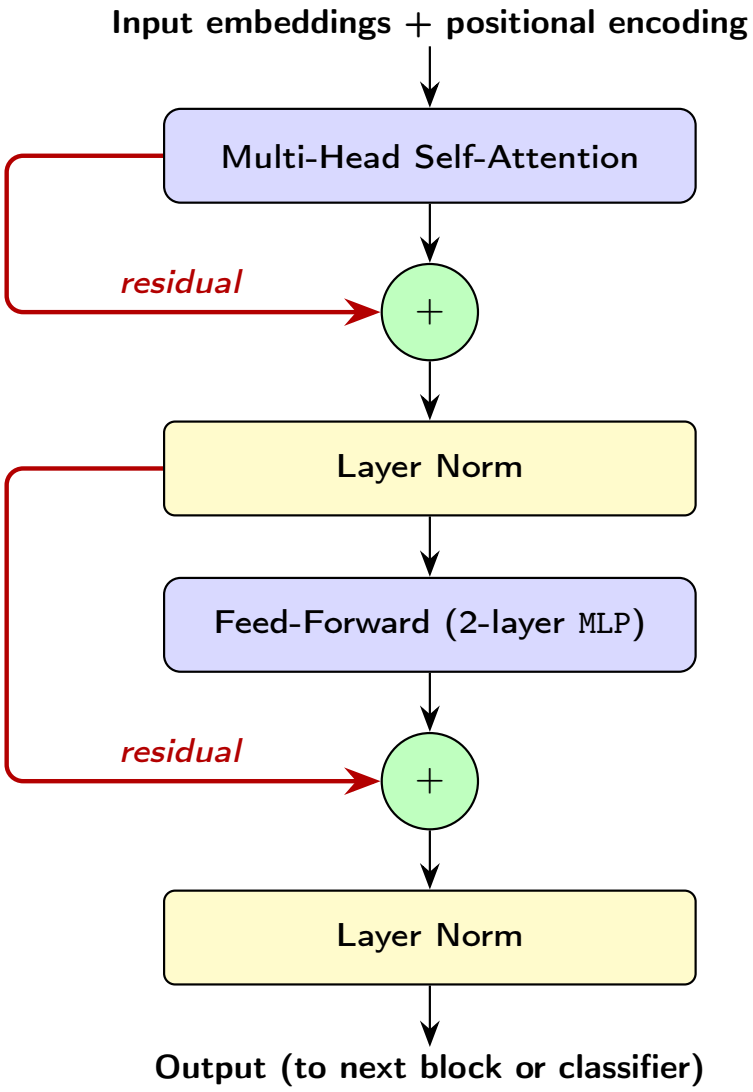
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Transformer Block: *Architecture* Diagram



Positional Encoding and Transformer Architecture

Positional Encoding

Self-attention is permutation-invariant; we must inject order information:

$$\text{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{2i/d}}\right), \quad \text{PE}(\text{pos}, 2i+1) = \cos\left(\frac{\text{pos}}{10000^{2i/d}}\right)$$

Transformer Block (stacked N times)

1. *Multi-Head Self-Attention* (with residual + LayerNorm)
2. *Position-wise Feed-Forward* (with residual + LayerNorm)

- Encoder-Decoder: original Transformer (*translation*)
- Encoder-only: BERT (*classification*); Decoder-only: GPT (*generation*)





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Transformer Block: Summary of Components

Why Transformers Win

1. No recurrence: compute all time steps in parallel (vs. sequential RNN)
2. Long-range interactions: direct connections between any two positions
3. Interpretable: attention weights show which tokens are *important*
4. Scalable: foundation for 100B+ parameter models (GPT-3, LLaMA)





Foundation Models: BERT and GPT

BERT (Devlin, 2019)

- Bidirectional *encoder*
- *Pretraining*: Masked LM + *Next Sentence Prediction*
- *Fine-tune* for: *classification*, *QA*, *NER*

GPT-2/3/4 (OpenAI)

- Autoregressive *decoder*
- *Pretraining*: *causal language modelling*
- *Few-shot / zero-shot* generation

Key Pattern

Pretrain on massive unlabelled data, then fine-tune on *small labelled sets* — the same paradigm as *Transfer Learning*.





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GPT-2/3/4 (OpenAI)

- Autoregressive *decoder*
- *Pretraining*: *causal language modelling*
- *Few-shot* / *zero-shot* generation

Key Pattern

Pretrain on massive unlabelled data, then fine-tune on *small labelled sets* — the same paradigm as *Transfer Learning*.





Practical Example: Sentiment Classification with Transformer

- **Dataset**: IMDB reviews ($50k$, *binary sentiment*)
- Model A: *2-layer* LSTM (baseline)
- Model B: small Transformer encoder (4 heads, 2 layers)
- Model C: pre-trained bert-base-uncased fine-tuned
- Metrics: *accuracy*, *precision*, *recall* on test set
- **Expected result**: BERT $\gg$ Transformer $\geq$ LSTM with the same compute budget





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Outline

1 Deep Learning Foundations

2 Convolutional Neural Networks

3 Sequence Models & Transformers

4 Unsupervised & Representation Learning





Unsupervised Learning Landscape

Unsupervised Learning Goal

Learn the underlying structure of **data** $\mathbf{x}$ without **labels** $\mathbf{y}$. Applications: *compression*, *feature discovery*, *anomaly detection*, *generation*.

- Density estimation: model the *data distribution* explicitly
- Dimensionality reduction: find compact representations (AE, PCA)





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AutoEncoder Architecture

Definition

A *neural network* trained to reconstruct its input through a compressed bottleneck (*latent code* $\mathbf{z}$):

$$\hat{\mathbf{x}} = \text{Dec}(\text{Enc}(\mathbf{x})), \qquad \mathcal{L}_{\text{AE}} = \|\mathbf{x} - \hat{\mathbf{x}}\|^2$$

- **Encoder**: $\mathbf{x} \to \mathbf{z}$; dimensionality reduction
- **Decoder**: $\mathbf{z} \to \hat{\mathbf{x}}$; reconstruction
- **Bottleneck**: forces the encoder to learn a compressed representation
- `Denoising AE`: corrupt input with *noise*; train to reconstruct clean — more robust features





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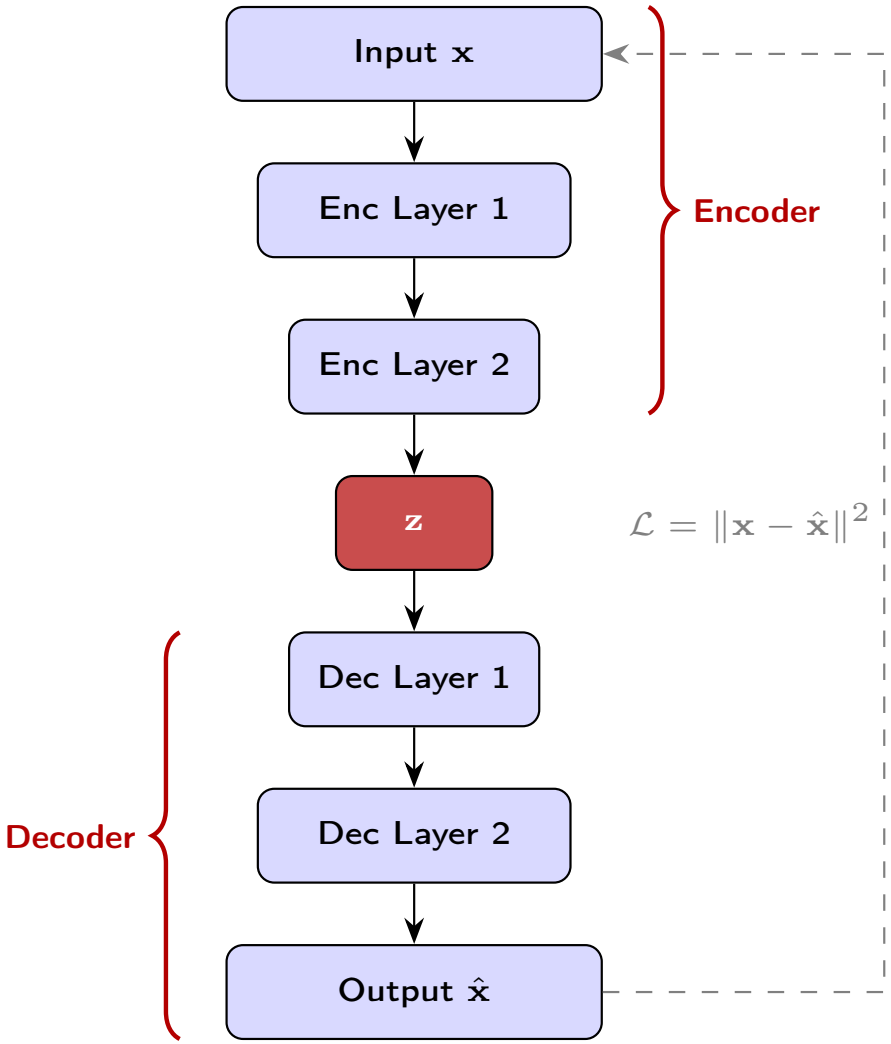
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AutoEncoder: Architecture Diagram



Latent Space and Anomaly Detection

Normal training

Train `AE` only on normal samples. The model learns to compress and reconstruct *typical patterns*.

`Anomaly` Scoring Algorithm

At *inference*, anomalous samples have high reconstruction error because the `encoder` was never trained to compress them:

$$\text{score}(\mathbf{x}) = \|\mathbf{x} - \text{Dec}(\text{Enc}(\mathbf{x}))\|^2$$

Threshold at the 95th *percentile* of training set errors.

- This is the core idea of *Challenge 6*
- Compare with `Isolation Forest` baseline





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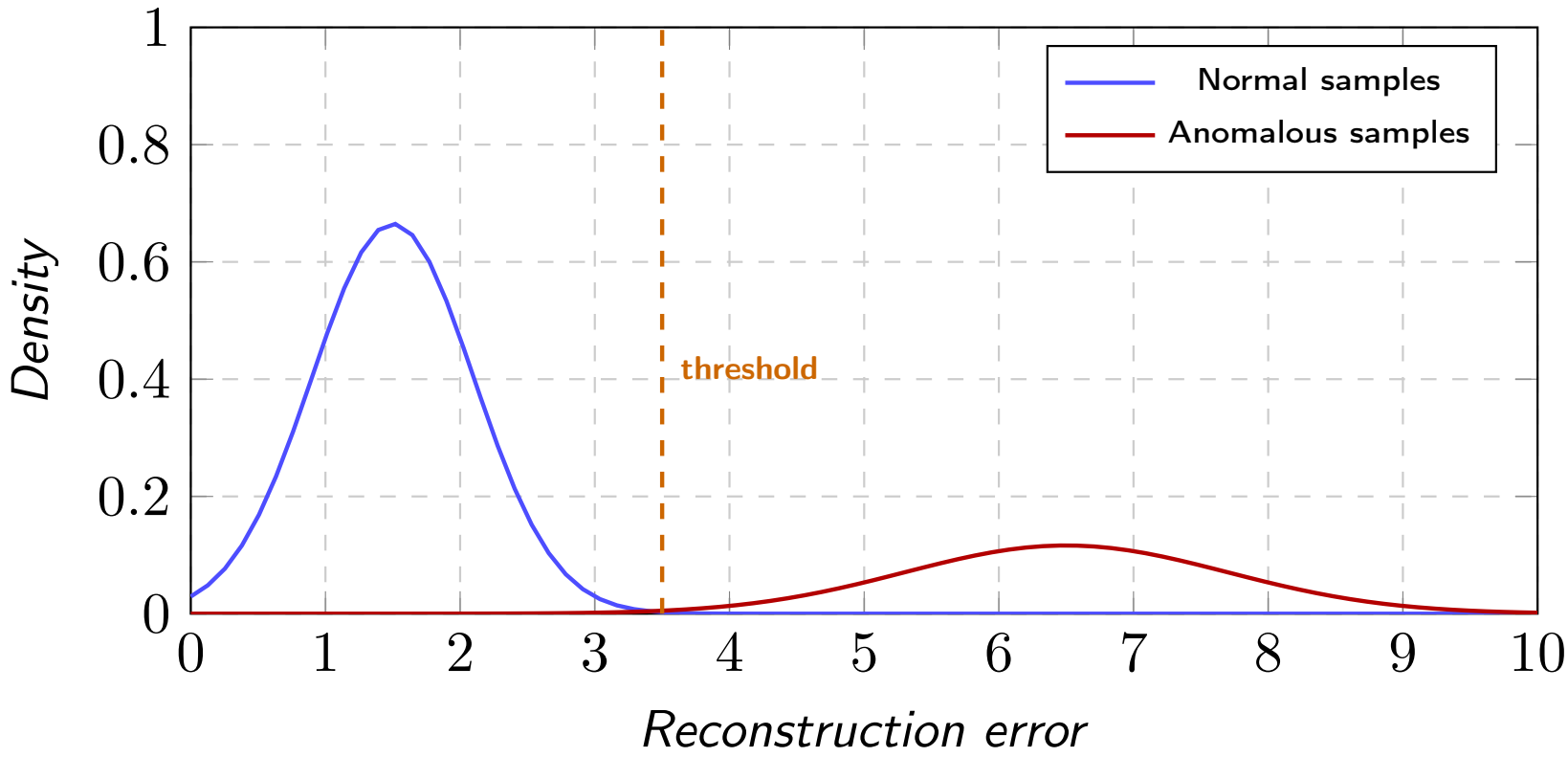
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Anomaly Detection: Reconstruction Error Distribution



Samples with error above the threshold are flagged as anomalies





Variational AutoEncoder (VAE)

Key Difference from AE (Kingma & Welling, 2014)

The encoder outputs a distribution over $\mathbf{z}$: $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$ rather than a single point $\mathbf{z}$.

ELBO Loss

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\textit{reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I}))}_{\textit{regularisation}}$$

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t-SNE and UMAP for Visualisation

`t-SNE` (van der Maaten, 2008)

- Converts *distances* to *conditional probabilities*
- Minimises `KL divergence` between high-D and 2-D distributions
- Preserves local structure
- Perplexity: neighbourhood size (5–50)
- Slow on large datasets

`UMAP` (McInnes, 2018)

- *Topological data analysis* foundations
- Preserves both local and global structure
- Much faster; scales to millions
- `n_neighbors`, `min_dist` control trade-off
- Preferred in practice

Visualisation Landscape

`PCA`: *fast, linear, preserves global structure*; `t-SNE`: *slower, nonlinear, preserves local structure*; `UMAP`: *balanced, preserves both local and global.*





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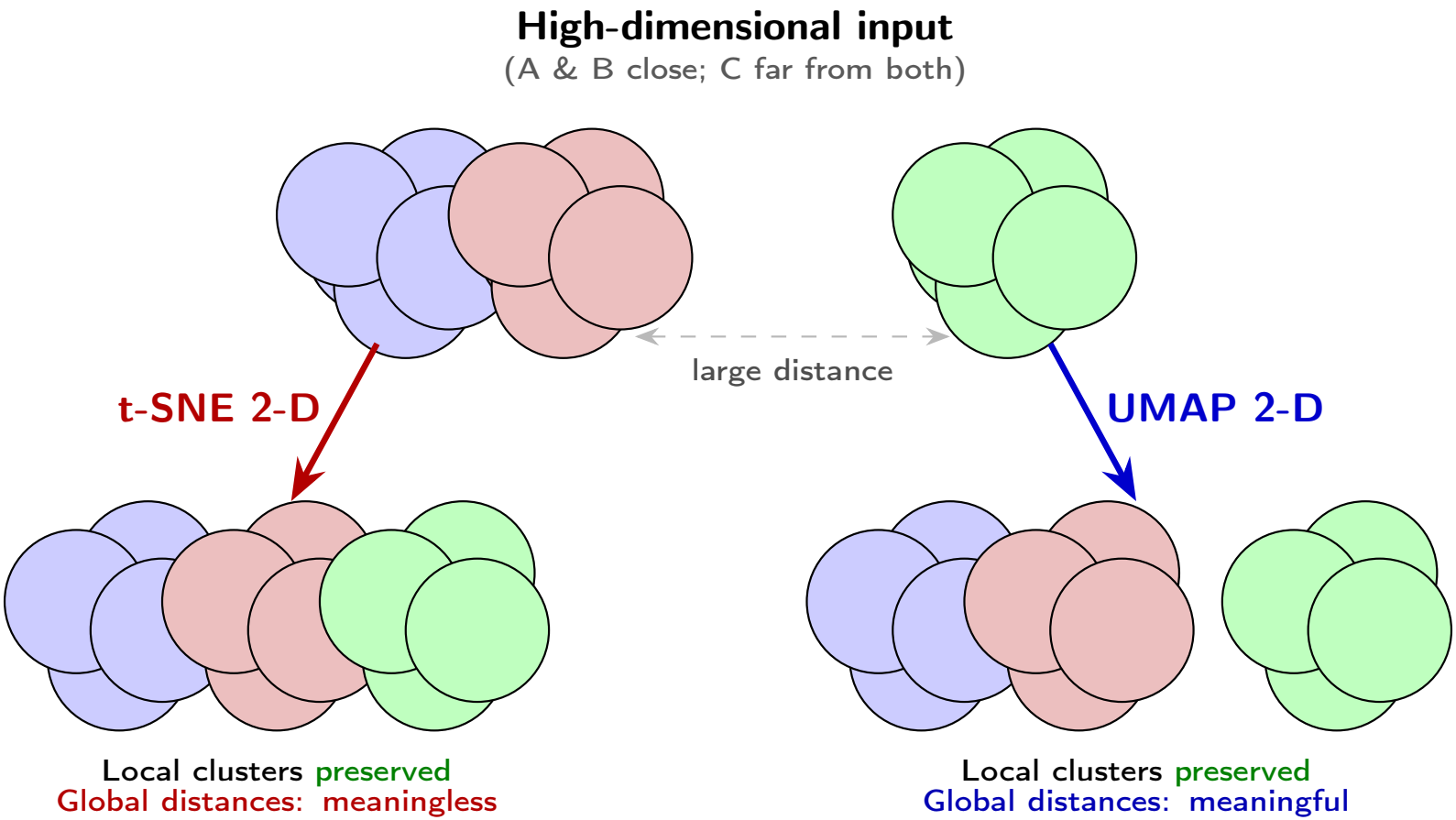
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t-SNE vs UMAP: Concept Diagram



Challenge 5 & 6 Connection

Challenge 5 — Clustering

Apply `K-Means`, `DBSCAN`, and `Hierarchical Clustering` to an assigned U.S. open government *dataset*. Visualise with `PCA` and `t-SNE/UMAP`.

Challenge 6 — `AutoEncoders` & Anomaly Detection

- Re-use the same preprocessed feature matrix from *Challenge 5*
- Train a standard `AE` and a `VAE`; compare *reconstruction error histograms*
- Visualise `VAE` *latent space* with `t-SNE` and `UMAP`
- Colour by *Challenge 5* cluster labels to compare paradigms
- Baseline: `Isolation Forest`; compare *Spearman rank correlation* with `AE` errors





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Practical Example: AE Anomaly Detection on Tabular Data

- **Dataset**: *Credit card fraud* (Kaggle), $284k$ transactions, 0.17% fraud
- **Protocol**: train AE on normal transactions only
- **Step 1**: *scatter plot* of *reconstruction errors* (*normal vs. fraud*)
- **Step 2**: threshold at 95th *percentile*; compute *precision*/*recall*
- **Step 3**: project *latent space* with UMAP; colour by *fraud label*
- **Expected**: fraud points cluster in high-error regions and appear as *outliers* in the 2-D projection





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Thanks!

Questions?



Repo: *https://github.com/EngAndres/ud-public/tree/main/courses/machine-learning*





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